

Mandatory Assignment 1

Development of Large Systems

This assignment is part of the final project.

Working on the semester project

Form a working group and register your team in the excel file on our OneDrive.

- Come up with your own project based on the final project requirements.
- Define the system requirements:
 - functional
 - non-functional requirements
- Choose a project management strategy for your project.
- Choose your technology stack.

Create a public GitHub repository for your project and provide the link in the excel file on our OneDrive.

System architecture design

- Draw the system architecture.
- Include the frontend application in the system architecture and describe how it interacts with backend services using REST and GraphQL.
- Choose proper communication channels between the microservices and the outside world.
- Define the architectural patterns that you will follow (for example CQRS, Snapshot pattern, etc.)
- Divide responsibilities within the team for the implementation of each part of the system.

Deployment strategy

- Investigate suitable cloud providers.
- Choose the cloud provider / providers for your system.
- Describe your CI/CD pipeline.
- Describe your initial testing strategy, including planned unit tests, integration tests, and system-level cooperation tests, and how these will be integrated into the CI/CD pipeline.
- Investigate the options for logging and monitoring of your system in production.

Documentation

Define documentation strategy.

Define versioning strategy: databases, APIs, code.

Report

Start writing the final project report (follow the required structure) - concentrate on the topics from this assignment.

Presentation

- Each working group will present their results during the class.
- Discussion
- Feedback - what to improve

Delivery to itsLearning before the deadline:

- Provide a link to your group GitHub monorepo
- Upload your PowerPoint presentation
- Report – covering this part